

CheckMate: English Grammatical Error Correction for Native Turkish Speakers

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Abstract—The covid-19 outbreak left many countries have no choice but turn to online education. Turkish students, who were among those who were affected, faced difficulties in improving their English as the opportunity to have face-to-face feedback was not available. In this work, we build and open-source a dataset for grammatical error correction composed of essays written by Turkish students from different universities capturing the errors Turkish native speakers tend to make. We utilize the dataset and build a model, which we deploy along with a web interface.

Keywords—Grammatical error correction, Data labeling, Data collection, Deep learning

I. INTRODUCTION

Grammar-checking tools have been utilized to assist the students who received online education during the Covid-19 outbreak to improve their writing skills. Turkish students studying in English were affected by the outbreak and missed the opportunity of getting quick face-to-face feedback.

Grammar error correction (GEC) aims to automatically correct various types of errors in the given text. Errors that violate the rules of English and the expected usage of English native speakers in morphological, lexical, syntactic, and semantic forms are all treated as a target to be corrected. Most GEC systems receive a raw, ungrammatical sentence as input, and return a refined, correct sentence as output.

The 2000s is the prior stage for GEC. In this stage, most of GEC systems are based on hand-craft rules incorporating the usage of parsers and linguistic characteristics. However, the complexity of designing rules and solving conflicts among the rules require a great magnitude of labor. Although some GEC works today still use rules as an additional source of correction, the performance of rule-based GEC systems has been superseded by data-driven approaches.

Remarkable progress has been achieved in GEC through data-driven approaches. In the late 2000s, classification-based approaches are developed to correct preposition errors and article errors. In this approach, classifiers are trained on a large magnitude of native error-free text to predict the correct target word, taking account into the linguistic features given by context. However, GEC systems that are capable of correcting all types of errors are more desirable. As an improvement, statistical machine translation (SMT) based GEC systems have gained significant attention in the early 2010s. In this approach, SMT models are trained on parallel sentence pairs, correcting all types of errors by “translating” the ungrammatical sentences

into refined output. More recently, with the increasing popularity of deep learning, neural machine translation (NMT) based GEC systems applying neural seq2seq models become dominant and achieved state-of-the-art performance. Since MT-based translation approaches require a corpus containing large parallel sentence pairs, language model-based GEC approaches have been utilized as an alternative that does not rely on supervised training.

In this work, we fill in the gap by introducing a grammar-checking tool for Turkish students who are learning English. To specialize the tool for Turkish students, we collected data from universities’ preparatory departments, annotated them, and utilized the annotated data to develop a grammar-checking tool. We open-source our dataset, our Python and Java annotation tools, and our pre-trained models. We also deploy our model and provide a web interface. In section II, we review the literature discussing previous related work. In section III, we describe our data collection, processing, and labeling process. And finally in IV and V, we describe the methodology and the results.

II. RELATED WORK

Computer Assisted Language Learning (CALL) [24] is a type of learning that combines foreign language learning and technology by making use of many fields such as psychology, sociology, natural language processing, linguistics, and artificial intelligence.

CALL has been a key element in the online education systems that were used during the Covid-19 break. Turkey was among the countries that decided to continue their education online through various platforms during the Covid-19 break, and this process affected especially new high school, preparatory, and freshmen students who received their education in English. The absence of possible face-to-face feedback in this situation urged the use of CALL.

Writing in a foreign language is an important skill for students to acquire in the preparatory and beginning years of their education. Since feedback is crucial for students to improve their writing skills, a computer-assisted online tool could fill in this gap.

The history of automatic proofreading programs dates back to the 1960s, the core functions of prominent programs such as Project Essay Grade [20] or The Writer’s Workbench [17] are still limited by misspellings and short and long sentences. It was known that there was a determination of the students,

and this showed the need for programs that provide more comprehensive and detailed feedback for those periods. For this reason, with the effect of technological developments, studies have accelerated in the last 20 years and advanced digital education tools and inspection programs have been created. Vantage Learning, Educational Testing Services, and Pearson Knowledge Technologies are leading companies in the field with programs that can provide detailed feedback to users with numerical values [8]. Programs that control discourse analysis, as well as time and cost advantages, have made significant advances in detailed feedback. MY Access designed by Vantage Learning is a widely used program in US public schools. With the artificial intelligence scoring engine called Intellimetric, it can analyze the writings of users in terms of semantic, syntactic, and discourse features [9]. Criterion, released by Educational Testing Services [26], is another web-based tool used by university students for exams such as GMAT and GRE [6]. The software engine of the tool, e-rater, scores and evaluates the articles by making comparisons with the articles controlled and scored by humans and machines in parallel with Intellimetric [1]. Thanks to natural language processing techniques, e-rater has a wide range of linguistic diagnoses in manuscripts, including grammar, language usage, style, or lexical complexity, thus providing feedback not only on form and accuracy but also on writing style and organization.

Grammar Error Correction (GEC) studies started with rule-based approaches. While the rule-based approach is simple, precise, and easy to interpret, such systems are very difficult to maintain and scale due to the abundance of individual errors and the complexity of the language. They also require language-specific knowledge and are not suitable for complex errors [11], [19]. Rules-based approaches are followed by data-driven approaches such as Language Model (LM)-based, Classification-based, Statistical Machine Translation (SMT), and Neural Machine Translation (NMT)-based approaches.

Language Model (LM) based approaches are easy to implement and can cover all error types compared to rule-based systems, they are also very flexible and only require a lot of unlabeled text. However, the approach has problems with rare or unseen words, as probability cannot cover all grammar, and LM-based approaches don't handle long-range dependencies very well. Creating a confusion set for error correction in this approach is another challenge. Error mode-based classifiers are trained using native, non-native, or hybrid data and are again more flexible than rule-based approaches. They are good for a certain set of error types [22]. Statistical Machine Translation-based systems can process not only words but also whole sentences. The SMT-based approach does not require any grammar to implement, but it does mean that a lot of parallel data actually needs to be trained. Such systems can fix any kind of error using the power of statistics. There is no feature engineering and the training phase is very simple if the required data is available. It is obvious that we will have a problem with words that are not yet in the vocabulary [3], [14], [18]. The Neural Machine Translation (NMT) based approach is the latest technology available in the field of grammar error correction. These models produce fluent sentences and correct all kinds of errors at the same time, such as the SMT-based approach. However, it requires large amounts of parallel training data and is computationally intensive and expensive,

making these models difficult to modify [2], [7], [12]. Data augmentation techniques such as noise injection and back translation can be applied to the datasets used in the creation of these models.

Existing automatic writing assessment programs can use dictionaries and sample writings thanks to their advanced technologies and provide progress reports for students and instructors. Given the number of students and the time required, automated spell-checking programs have become essential, especially during the COVID-19 era, to ease the burden of instructors and provide comprehensive feedback to students. Although the programs designed so far (such as Criterion and Grammarly) provide advanced feedback as desired, they cannot be used for the desired purposes in our country because they do not offer completely free use and are not fully suitable for the needs of Turkish students learning English. Based on the studies on the common writing mistakes of Turkish students studying English [10], [15], [16], [23], the most common mistakes made by Turkish students are pluralization, prepositions, pronoun and verb tenses. These studies have shown that existing programs fail to detect these particular errors (eg, fail to detect pluralization and verb tense errors) or perform poorly in detecting such errors (Criterion's accuracy rate is around 70% for prepositions and pronouns).

III. DATA PREPARATION

A. Data Collection & Processing

We conducted meetings with representatives from the English preparatory department of Ozyegin, Bogazici, and Nisantasi University. In those meetings, we discussed the level and the format of the essays to be collected. We acquired essays either handwritten or digitally written. The digitally written ones came in either PDF or Word format.

We start the processing phase of the data by removing anything other than the body of the essay. And in parallel, we spend some time digitizing the handwritten essays. We then tokenize all the essays into sentences.

B. Data Labelling

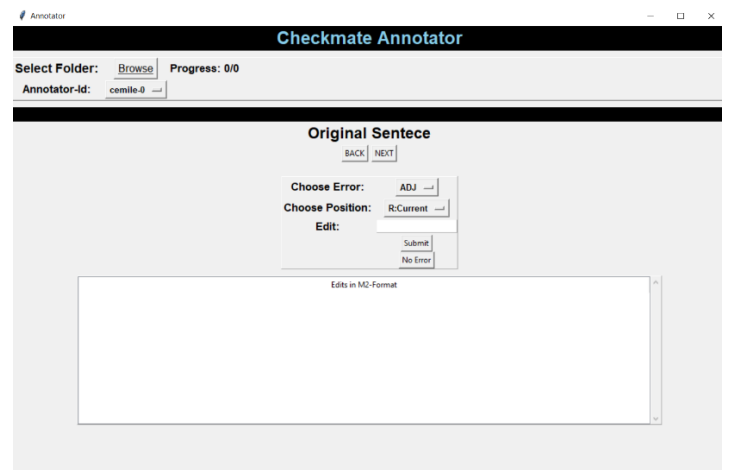


Figure 1: An annotation tool developed with Python

We follow the M2 format to annotate the data. In M2 format¹, a line preceded by S denotes an original sentence while a line preceded by A indicates an edit annotation. Each edit line consists of the start and end token offset of the edit, the error type, and the tokenized correction string. The next two fields are included for historical reasons and can be ignored, while the last field is the annotator id. A noop edit is a special kind of edit that explicitly indicates an annotator/system made no changes to the original sentence. If there is only one annotator, noop edits are optional, otherwise a noop edit should be included whenever at least 1 out of n annotators considered the original sentence to be correct. This is something to be aware of when combining individual M2 files, as missing noops can affect results. To annotate the data with the above-mentioned format, we built an interface for the annotators that is easy to use. In order to get the annotators to start as soon as possible and also get used to the process, we quickly built an interface using Python’s Tkinter. Figure 1 shows our Python annotation tool. With this simple annotation tool, we introduced the problem to the annotators. And in the meantime, we were working on a more efficient and faster annotation tool in Java², which we show in Figure 2. The tool takes as input a path to a folder containing files each representing a sentence formatted in a specific format. Each annotation information of a sentence is registered into the corresponding file of that sentence.

Because the data collection phase took months, we started with other phases of the project as we were receiving the data. Therefore, all the sentences were not annotated in one batch. Besides, because the handwritten essays came late and digitizing them took some time, they haven’t been annotated yet.

The data was annotated in 3 batches³. The first batch was annotated with the Python annotation tool, and the second and the third batches with the Java interface. We combined them all later in one place. We had two annotators, and each annotator annotated the same sentences in batches 2 and 3. Batch 1 on the other hand was different for each annotator. Annotator X annotated 4956 sentences, while annotator Y annotated 4473 sentences. We show in Table I the error types found in the data and their frequencies both by annotator X and annotator Y.

IV. METHODS

A. Rule-Based Methods

We mention here some of the rules our linguists hand-crafted. The rules are involved in cases such as superlative and comparative adjectives, singular nouns, and determiners. Some of the rules detect the occurrence of an error such as checking whether a superlative adjective is preceded by a determiner. Other rules detect and correct the error such as the one that checks the correct usage of a & an.

Our system is not using those rules yet because of their small number, only 10 rules. Also, they are still being tested and improved. Once more rules are implemented and tested,

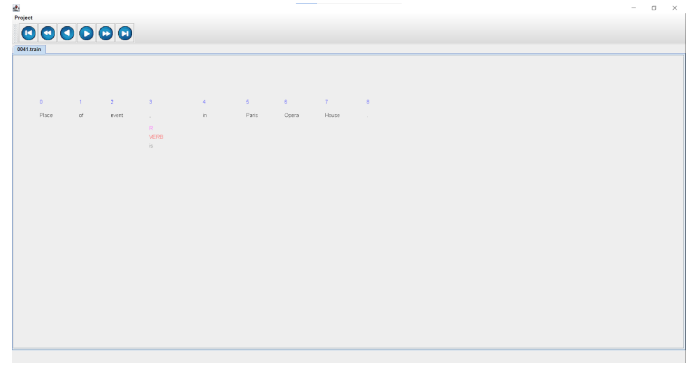


Figure 2: An annotation tool developed with Java

Error Type	Freq. (checkMate-X)	Freq. (checkMate-Y)
ADJ	0.0432	0.0478
PRON	0.0315	0.0349
DET	0.0535	0.0592
NOUN:NUM	0.0385	0.0427
CONJ	0.0323	0.0358
VERB:FORM	0.0067	0.0074
PREP	0.0682	0.0756
VERB	0.0914	0.1013
NOUN	0.0533	0.0590
PART	0.0246	0.0273
PUNCT	0.0230	0.0255
VERB:INFL	0.0365	0.0405
VERB:TENSE	0.0117	0.0130
NOUN:INFL	0.0004	0.0004
ADV	0.0085	0.0094
OTHER	0.0006	0.0007
SPELL	0.0291	0.0322
UNK	0.0042	0.0047
None	0.0048	0.0054
ORTH	0.0129	0.0143
ADJ:FORM	0.0050	0.0056
CONTR	0.0000	0.0000
MORPH	0.0002	0.0002
VERB:SVA	0.0216	0.0239
NOUN:POSS	0.0000	0.0000

Table I: Error Types and their percentages in both checkMate-X and checkMate-Y datasets.

we could compare a rule-based system with a neural system. Besides, we could incorporate the rules into a neural system to work together.

B. Neural Methods

We take advantage of Google’s pre-trained model Text-To-Text Transfer Transformer (T5) [21]. The T5 model’s architecture is based on the Transformer architecture, which was first introduced in [25]. The architecture consists of a stack of encoder and decoder layers, each layer consisting of a multi-head self-attention sublayer followed by a feed-forward network sublayer.

The encoder layers process and encode the input sequence into a context representation. The decoder layers then use this context representation to generate the output sequence.

The T5 model is trained on several datasets for 18 different tasks such as Text Summarization, Question Answering, and Machine Translation. T5 has been utilized for the Grammar Error Correction task as well. Similarly, we use T5 and fine-tune it on our dataset.

¹<https://github.com/chrisjbryant/errant>

²<https://github.com/StarlangSoftware/GrammaticalErrorCorrection>

³<https://github.com/asimokby/checkmate-gec>

Training Set	Test Set	TP	FP	FN	Precision	Recall	F0.5
checkMate-X	checkMate-X dev	157	436	498	26.48	23.97	25.93
checkMate-Y	checkMate-Y dev	185	500	574	27.01	24.37	26.44
W&I+LOCNESS	W&I+LOCNESS dev	4938	8200	9984	37.59	33.09	36.59
W&I+LOCNESS + checkMate-X	W&I+LOCNESS dev	4474	6790	10448	39.72	29.98	37.30
W&I+LOCNESS + checkMate-Y	W&I+LOCNESS dev	4206	9938	10716	29.74	28.19	29.41

Table II: Results of the fine-tuned T5 model. checkMate-X and checkMate-Y are the two datasets we introduce each being annotated by a different annotator. The precision, recall, and F0.5 are shown for each experiment.

V. RESULTS

In our experiments, we do not combine what annotators X and Y annotated and hence we have two different datasets which we will call here checkMate-X and checkMate-Y. We split each dataset into train and test sets. We fine-tune T5 on our training set and use the test set for evaluation. We use ERRANT [5], which is a toolkit used by the NLP community for producing error types scores for the Grammatical Error Correction (GEC) task. Furthermore, we utilize the dataset W&I+LOCNESS [4], [13], which contains 77384 sentences, and combine it with our data, which we detail in II. Our results show that combining our checkMate-X dataset with the W&I+LOCNESS dataset does improve the performance by .71%. However, surprisingly, combining W&I+LOCNESS with checkMate-Y causes the F0.5 score to significantly decrease.

VI. CONCLUSION

As online education emerged in the past couple of years due to the Covid-19 outbreak, Turkish students missed the opportunity to have face-to-face and quick feedback. In this work, we fill in the gap by building a grammatical error correction dataset capturing mistakes Turkish students tend to make when writing in English. We develop rules for a rule-based system and also fine-tune a pre-trained model that we deploy a model along with a web interface. Future work could merge our two datasets checkMate-X and checkMate-Y into a single dataset. Furthermore, investigate the dataset more to highlight the errors which are common for Turkish students.

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